

6.02 Work, Energy and Power			
6.02a 6.02c 6.02b	Work	a) Understand the concept of work done by a force. b) Be able to calculate the work done by a constant force. <i>The force may not act in the direction of motion of the body and so learners will be expected to resolve forces in two dimensions.</i>	c) Be able to calculate the work done by a constant force in two dimensions using vectors ($\mathbf{F} \cdot \mathbf{x}$) or by a variable force ($\int F dx$) in one dimension only.
6.02d 6.02f 6.02e	Energy	d) Understand the concept of the mechanical energy of a body. <i>i.e. The kinetic and potential energy.</i> e) Be able to calculate the gravitational potential energy (mgh) and kinetic energy ($\frac{1}{2}mv^2$) of a body.	f) Be able to calculate the kinetic energy of a body using the scalar product $\frac{1}{2}m\mathbf{v} \cdot \mathbf{v}$ <i>Learners may be expected to use the formula $\mathbf{v} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{u} + 2\mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{v}$ in solving a variety of problems, for example in calculating kinetic energy of a body.</i>

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
6.02g 6.02h	Hooke's law		g) Understand and be able to use Hooke's law, in the form $T = \frac{\lambda x}{l}$, for elastic strings and springs. <i>Includes an informal understanding of when Hooke's law does not apply.</i> h) Be able to calculate the elastic potential energy $\left(E = \frac{\lambda x^2}{2l} \right)$ stored in a string or spring. <i>Learners will be expected to state the formula for the elastic potential energy stored in a string or spring unless they are explicitly asked to derive it.</i>
6.02i 6.02j	Conservation of energy	i) Understand and be able to use the principle of the conservation of mechanical energy and the work-energy principle for dynamic systems, including consideration of energy loss.	j) Extend their knowledge of the principle of the conservation of mechanical energy and the work-energy principle to systems which include elastic strings or springs.
6.02k 6.02l 6.02m	Power	k) Understand and be able to use the definition of power (the rate at which a force does work). <i>Includes average power = $\frac{\text{work done}}{\text{time elapsed}}$.</i> l) Be able to use the relationship between power, the tractive force and velocity ($P = Fv$) to solve problems. <i>e.g. Motion on an inclined plane.</i> <i>Includes maximum velocity and speed.</i> <i>Learners will be required to resolve forces in two dimensions.</i>	m) Be able to calculate the power associated with a variable force in two dimensions using the scalar product $P = \mathbf{F} \cdot \mathbf{v}$